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Particle filter divergence monitoring with application to terrain navigation

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Abstract—Particle filters are an efficient Monte-Carlo method for Bayesian estimation in non-linear models. However, under certain circumstances, they are subject to divergence. Increasing the number of particles is not always possible so it is essential for many applications to assess the reliability of the solution provided by the filter. In terrain navigation, trusting an erroneous position estimate can be problematic for obvious reasons. We introduce a framework for detecting filter divergence in the case of scalar measurements. The detector is based on a sequential change detection algorithm and we illustrate its performance on several terrain navigation scenarios.

I. INTRODUCTION

In positioning applications, we are interested in estimating a given state (position, velocity...) partially observed through measurements obtained via external sensors. The estimation problem is typically solved by computing the filter i.e the posterior $p_k = p(x_k|y_{0:k})$ where x_k is the unobserved state and $y_{0:k} = \{y_0, y_1, \dots, y_k\}$ is the set of measurements up to the current time k . In Bayesian estimation, the filter is obtained sequentially through prediction and measurement update steps. When dealing with non-linear systems, a closed form of the filter is not available in general so it is necessary to resort to approximation schemes such as Kalman filters, grid-based filters, or more frequently particle filters [1]. The latter can, in theory, approximate any probability distribution by sampling N particles from a prior distribution and weighting them according to the observations. The *SIR (Sampling Importance Resampling)* is the usual framework for computing iteratively the particle approximation and consists in :

- a *prediction step* where the samples are propagated according to dynamical model.
- a *weight update step* where each sample is weighted proportionally to its likelihood given the current measurement.
- a *resampling step* which duplicates particles with a high weight/discards particles with a low weight.

In practice, the use of a finite amount of particles does not guarantee that our filter will converge every time. From time to time, divergent tracks will appear and it is crucial for safety critical applications such as terrain aided navigation [2] to assess the reliability of the particle estimate. A divergence occurs when the particle cloud deviates noticeably from the

true state such that it cannot recover the right track. The filter usually has to be restarted either from the last assumed reliable estimate or from the current state.

The causes of filter divergence can be attributed to modelling errors or to the algorithm itself. In the first case, the dynamical model may not be accurate enough to describe the true state trajectory. However, as far as terrain navigation is concerned, this is not an issue. Errors in the observation model may also cause the filter to diverge since they result in a wrong likelihood evaluation. For instance, a radar-altimeter doesn't always measure the ground clearance when navigating along a very rough terrain. It has a tendency to measure the shortest distance to the peak covered by the beam lobe.

Divergent tracks can also be caused by the filter design and/or by the values of the hyperparameters of the model. For example, when the observation function is ambiguous, the particle approximation is multimodal and it has been noted that the resampling step can lead to a loss of the correct mode ([3], [4]). Moreover, if the measurement noise is additive with small variance σ_v , most particles tend to have a very small weight when the sampling density is the prior $p(x_k|x_{k-1})$. Indeed, the likelihood is so peaked that it may have little overlap with the prior. This usually leads to a high divergence rate as the resampling step is not enough to put the particles closer to the region of interest.

The literature on divergence monitoring is relatively scarce to our knowledge. In the field of indoor localization [5], abrupt changes in the target's motion may cause the filter to make incorrect estimates because of the inability of the dynamical model to handle non-smooth trajectories. The ensuing divergence is detected by computing the Kullback-Leibler divergence between the probability surfaces associated with the current observation and the particle cloud. In [6], the unnormalized importance weights are used as a means of detection. Similarly, in [7], the mean of the unnormalized weights is used as an indicator of filter degeneracy.

In this article, we will assume that the dynamical and observation models are accurate enough and focus only on divergent tracks that are caused by the filtering algorithm itself. The main contribution of this paper consists in coupling the non-linear filter with a change detection algorithm based on the innovation sequence in order to quickly detect the filter

divergence.

The paper is organized as follows. The general framework of the filter divergence detection is detailed in section II. Section III is devoted to the change detection algorithm. The performances of this sequential change detection test coupled with a particle filter is studied in section IV. Finally, some conclusions are drawn in section V.

II. A GENERAL FRAMEWORK FOR DIVERGENCE DETECTION OF NON-LINEAR FILTERS

Consider the following state-space model

$$\begin{cases} x_k = f(x_{k-1}) + w_k \\ y_k = h(x_k) + v_k \end{cases} \quad (1)$$

where the process noise w_k and the measurement noise v_k are mutually independent, i.i.d sequences. We assume that $w_k \sim \mathcal{N}(0, Q)$ and $v_k \sim \mathcal{N}(0, R)$. The state vector x_k is assumed to lie in $\mathbb{R}^d$.

A filter provides an estimate at each time step $\hat{x}_k$ and we wish to detect the instant where $\hat{x}_k$ gets too far from the unknown true state given the associated uncertainty $\hat{\Sigma}_k = \text{Cov}(\hat{x}_k)$.

A particle filter approximates $p(x_k|y_{0:k})$ by a Dirac mixture through the following steps (SIR with prior proposal [8]) :

- 1) **Initialisation** : sample $x_0^i \sim p(x_0)$, $i = 1 \dots N$ and set $\omega_0^i = \frac{1}{N}$
- 2) **Prediction** : sample $x_k^i \sim p(x_k|x_{k-1}^i)$, $i = 1 \dots N$
- 3) **Measurement update** :
 - compute the weights $\tilde{\omega}_k^i = \omega_{k-1}^i g(y_k|x_k^i)$ where g is the likelihood function
 - set $\omega_k^i = \frac{\tilde{\omega}_k^i}{\sum_{j=1}^N \tilde{\omega}_k^j}$
 - compute $\hat{x}_k = \sum_{i=1}^N \omega_k^i x_k^i$
- 4) **Resampling step** : discard/multiply particles $\{x_k^i\}$ according to high/low weights ω_k^i

A. The case of the Kalman filter

If the above system is linear, the optimal filter is the Kalman filter. It is well known in the literature [9] that wrong values for Q and R may cause the predicted state covariance to be quite small compared to the actual error which is a sign of divergence.

In the Kalman filter literature, the innovation is defined as

$$\epsilon_k = y_k - \hat{y}_{k|k-1}$$

It reflects the discrepancy between the received measurement y_k and the predicted measurement $\hat{y}_{k|k-1} = \mathbb{E}(y_k|y_{0:k-1})$. ϵ_k is an i.i.d zero mean Gaussian process [10] with covariance matrix $\Sigma_k = H P_{k|k-1} H^T + R$.

Let H_0 be the hypothesis of normal filter behavior. Let $\eta_k = \epsilon_k^T \Sigma_k^{-1} \epsilon_k$. Given H_0 , η_k follows a chi-square distribution χ_2^d with d degrees of freedom. Divergence can be declared whenever $\eta_k > q_\alpha$ where the threshold q_α is such that $\mathbb{P}(\eta_k > q_\alpha | H_0) < \alpha$.

B. Hypothesis testing for divergence detection in the non-linear case

Given the previous result, one would like to derive a statistics based on the innovation sequence in the general case where either f or h are non-linear assuming that a particle filter is used.

In the sequel, ϵ_k^N stands for the innovation of the particle filter. Determining the probability distribution of ϵ_k^N seems difficult in this case, however we can derive the mean and approximate the variance $\sigma_\epsilon^2(k)$.

Throughout the rest of the article we will assume the measurements y_k to be scalar and we will consider the null hypothesis H_0 : 'the filter has not diverged'.

The mean of ϵ_k is always zero :

$$\mathbb{E}(\epsilon_k | y_{0:k-1}) = 0 \quad (2)$$

$$\begin{aligned} \epsilon_k^N &= y_k - \hat{y}_{k|k-1}^N \\ &= y_k - \hat{y}_{k|k-1} + \hat{y}_{k|k-1} - \hat{y}_{k|k-1}^N \\ &= \epsilon_k + \hat{y}_{k|k-1} - \hat{y}_{k|k-1}^N \end{aligned} \quad (3)$$

where

$$\hat{y}_{k|k-1} = \mathbb{E}(y_k | y_{0:k-1}) = \int h(x_k) p(x_k | y_{0:k-1}) dx_k$$

is the observation predicted by the optimal filter and

$$\hat{y}_{k|k-1}^N = \sum_{i=1}^N \omega_{k-1}^i h(x_k^i)$$

is its particle counterpart.

Let

$$\zeta_k^N = \hat{y}_{k|k-1} - \hat{y}_{k|k-1}^N$$

Then given hypothesis H_0 , as N grows to infinity $\mathbb{E}(\zeta_k^N) \rightarrow 0$. Finally,

$$\mathbb{E}_{H_0}(\epsilon_k^N | y_{0:k-1}) \approx 0 \quad (4)$$

Now consider $\mathcal{Y}^{k-1} = \text{span}(y_0, \dots, y_{k-1})$. $\hat{y}_{k|k-1}$ can be viewed as the projection of the current observation on $\mathcal{Y}^{k-1}$ hence $\hat{y}_{k|k-1} \in \mathcal{Y}^{k-1}$. $\hat{y}_{k|k-1}^N$ only depends on measurements up to time $k-1$ so $\hat{y}_{k|k-1}^N \in \mathcal{Y}^{k-1}$. This entails that $\zeta_k^N \in \mathcal{Y}^{k-1}$. ϵ_k is the innovation so it is orthogonal to $\mathcal{Y}^{k-1}$. This implies that ϵ_k and ζ_k^N are uncorrelated.

The variance of the particle filter innovation can then be worked out :

$$\sigma_\epsilon^2(k) = \text{Var}(\epsilon_k^N | \mathcal{Y}^{k-1}) \quad (5)$$

$$= \text{Var}(\epsilon_k + \zeta_k^N | \mathcal{Y}^{k-1}) \quad (6)$$

$$= \text{Var}(\epsilon_k | \mathcal{Y}^{k-1}) + \text{Var}(\zeta_k^N | \mathcal{Y}^{k-1}) \quad (7)$$

$$\begin{aligned} \text{Var}(\epsilon_k | \mathcal{Y}^{k-1}) &= \text{Var}(v_k + h(x_k) - \mathbb{E}(h(x_k) | y_{0:k-1}) | \mathcal{Y}^{k-1}) \\ &= \sigma_v^2 + \text{Var}(h(x_k) - \mathbb{E}(h(x_k) | y_{0:k-1}) | \mathcal{Y}^{k-1}) \\ &= \sigma_v^2 + \text{Var}(h(x_k) | \mathcal{Y}^{k-1}) \end{aligned}$$

To compute $\text{Var}(\zeta_k^N | \mathcal{Y}^{k-1})$, notice that

$$\begin{aligned}\hat{y}_{k|k-1} &= \int h(x_k) p(x_k | y_{0:k-1}) dx_k \\ &= \iint h(x_k) p(x_k | x_{k-1}) p(x_{k-1} | y_{0:k-1}) dx_k dx_{k-1} \\ &= \int \psi(x_{k-1}) p(x_{k-1} | y_{0:k-1}) dx_{k-1}\end{aligned}$$

where $\psi(x_{k-1}) = \int h(x_k) p(x_k | x_{k-1}) dx_k$

Thus,

$$\begin{aligned}\zeta_k^N &= \hat{y}_{k|k-1} - \hat{y}_{k|k-1}^N \\ &= \int h(x_k) p(x_k | y_{0:k-1}) dx_k - \sum_{i=1}^N \omega_{k-1}^i h(x_k^i) \\ &= \int \psi(x_{k-1}) p(x_{k-1} | y_{0:k-1}) dx_{k-1} - \sum_{i=1}^N \omega_{k-1}^i \psi(x_{k-1}^i)\end{aligned}$$

The central limit theorem (CLT) [11] states that when the SIS¹ algorithm is used, for any bounded function ϕ :

$$\sqrt{N} \left(\int \phi(x_k) p(x_k | y_{0:k}) dx_k - \sum_{i=1}^N \omega_k^i \phi(x_k^i) \right) \xrightarrow{\mathcal{D}} \mathcal{N}(0, v_k(\phi))$$

where

$$v_k(\phi) = \frac{\int (\phi(x_k) - \bar{\phi}_k)^2 g^2(y_{0:k} | x_{0:k}) p(x_{0:k}) dx_{0:k}}{\left(\int g(y_{0:k} | x_{0:k}) p(x_{0:k}) dx_{0:k} \right)^2} \quad (8)$$

g being the likelihood function and $\bar{\phi}_k = \int \phi(x_k) p(x_k | y_{0:k}) dx_k$.

Under H_0 , if N is large enough, $\text{Var}(\zeta_k^N)$ can be approximated by considering $\phi = \psi$ in the CLT :

$$\text{Var}_{H_0}(\zeta_k^N | \mathcal{Y}^{k-1}) \approx \frac{v_{k-1}(\psi)}{N} \quad (9)$$

1) *Approximation of $v_{k-1}(\psi)$* : We compute $v_{k-1}(\psi)$ via the particle approximation as in [12]. Because we perform resampling occasionally, $v_{k-1}(\psi)$ cannot be computed directly via the CLT. However, the SIS can be viewed as an instance of the SIS between two successive resampling steps. Therefore, $v_{k-1}(\psi)$ can be computed from (8) by accumulating observations between two resampling steps. Before the first resampling step,

$$v_{k-1}(\psi) \approx \frac{N \sum_{i=1}^N (\psi(x_{k-1}^i) - \bar{\psi}_{k-1})^2 g^2(y_{0:k-1} | x_{0:k-1}^i)}{\left(\sum_{i=1}^N g(y_{0:k-1} | x_{0:k-1}^i) \right)^2}$$

Recall

$$\begin{aligned}\psi(x_{k-1}^i) &= \int h(x_k) \overbrace{p(x_k | x_{k-1}^i)}^{\approx \delta(f(x_{k-1}^i))} dx_k \\ &\approx h(f(x_{k-1}^i)) \\ &= h(x_{k-1}^i)\end{aligned}$$

¹Sequential Importance Sampling : it is analogous to the SIS without the resampling step

In the previous integral, the prediction density $p(x_k | x_{k-1}^i)$ is approximated by a Dirac around the predicted mean x_k^i . For our application, this is sufficient since the noise covariance Q is quite small (see IV).

We also note that $\bar{\psi}_{k-1} \approx \sum_{i=1}^N \omega_{k-1}^i \psi(x_{k-1}^i)$

Finally,

$$v_{k-1}(\psi) \approx \frac{N \sum_{i=1}^N (h(x_{k-1}^i) - \bar{\psi}_{k-1})^2 g^2(y_{0:k-1} | x_{0:k-1}^i)}{\left(\sum_{i=1}^N g(y_{0:k-1} | x_{0:k-1}^i) \right)^2} \quad (10)$$

If k is after the first resampling occurrence,

$$v_{k-1}(\psi) \approx \frac{N \sum_{i=1}^N (h(x_{k-1}^i) - \bar{\psi}_{k-1})^2 g^2(y_{0:k-1} | x_{0:k-1}^i)}{\left(\sum_{i=1}^N g(y_{0:k-1} | x_{0:k-1}^i) \right)^2} \quad (11)$$

k_0 being the last resampling time before k .

2) *Approximation of $\text{Var}(h(x_k) | y_{0:k-1})$* : Using the particle system, we obtain

$$\text{Var}(h(x_k) | y_{0:k-1}) \approx \sum_{i=1}^N w_{k-1}^i (h(x_k^i) - \bar{h}_{k|k-1})^2$$

where $\bar{h}_{k|k-1} = \sum_{i=1}^N w_{k-1}^i h(x_k^i)$.

Given the previous results, we may define the null hypothesis for the innovation sequence as

$$\begin{cases} \mathbb{E}_{H_0}(\epsilon_k^N | \mathcal{Y}^{k-1}) = 0 \\ \text{Var}_{H_0}(\epsilon_k^N | \mathcal{Y}^{k-1}) = \sigma_\epsilon^2(k) \end{cases} \quad (12)$$

where

$$\sigma_\epsilon^2(k) \approx \sigma_v^2 + \sum_{i=1}^N w_{k-1}^i (h(x_k^i) - \bar{h}_{k|k-1})^2 + \frac{v_{k-1}(\psi)}{N}$$

A test for detecting abnormal mean shifts can then be used to monitor the filter's behavior.

The first step is to consider the normalized residual $r_k^N = \frac{\epsilon_k^N}{\sigma_\epsilon(k)}$ which is zero mean and has unit variance under the null hypothesis H_0 .

The most straightforward solution would be to test $\mathbb{E}(r) = 0$ against $\mathbb{E}(r) \neq 0$. However, since the whole distribution of r is not available, methods such as the Generalized Likelihood Ratio cannot be applied to this test. For this reason, we formulate the alternative hypothesis as a minimum change in the mean value of r_k i.e.:

$$\begin{cases} |\mathbb{E}_{H_1}(r_k^N | \mathcal{Y}^{k-1})| \geq \nu \\ \text{Var}_{H_1}(r_k^N | \mathcal{Y}^{k-1}) = 1 \end{cases} \quad (13)$$

The choice of ν will be addressed in the following section.

III. CHANGE DETECTION ALGORITHMS

A. The CUSUM algorithm

To solve the problem of filter divergence detection, it is necessary to design a sequential test based on the innovation sequence which is able to detect any divergent track as

soon as possible. The *CUSUM* algorithm [13] is a suitable candidate. It is also reported to perform well in the presence of non Gaussian innovations which makes it interesting to the considered application.

Let $u_1, u_2, \dots, u_k$ be a Gaussian random sequence such that

$$\begin{cases} \text{Var}(u_k) = \sigma^2 & \forall k \\ m \stackrel{\text{def}}{=} \mathbb{E}(u_k) = 0 & \text{for } k \leq t_0 \\ m = \nu & \text{for } k \geq t_0 \end{cases} \quad (14)$$

The goal of the *CUSUM* test is to detect the change point t_0 as soon as possible provided that the mean time before a false alarm is bounded from below by a given constant. Let $\mathbb{E}_{t_0}^1$ be the expectation with respect to the probability measure $P_{t_0}^1$, where the change point is equal to t_0 . Denote t_a the stopping time of the test. The Lorden's criterion of optimality is the following (see details in [14]). We require that the worst case mean detection delay ²

$$\bar{\mathbb{E}}^*(t_a) = \sup_{t_0 \geq 1} \text{esssup} \mathbb{E}_{t_0}^1 [(t_a - t_0 + 1)^+ | \mathcal{U}_1^{t_0-1}], \quad (15)$$

where $\mathcal{U}_1^{t_0-1} = (u_1, \dots, u_{t_0-1})$, $x^+ = \max(0, x)$, should be as small as possible for a given mean time before a false alarm γ

$$\mathbb{E}_0(t_a) \geq \gamma \quad (16)$$

The *CUSUM* test does in fact minimize the worst case mean detection delay. The stopping time (alarm time) of the *CUSUM* test is defined as follows :

$$\begin{aligned} t_a &= \inf \{k \geq 1 : g_k \geq h\} \\ g_k &= \left(g_{k-1} + \frac{\nu}{\sigma^2} \left(u_k - \frac{\nu}{2} \right) \right)^+ \end{aligned} \quad (17)$$

where $g_0 = 0$ and h is a threshold. The tuning parameters h and ν define a tradeoff between the mean time between 2 false alarms and the detection delay also taking into account the sensitivity of the *CUSUM* with respect to the change amplitude ν .

In our case the innovations are not necessarily *i.i.d* Gaussian before and after change. We only use the previously given assumptions to numerically define the performances of the *CUSUM* test as a preliminary approximation.

Because the sign of the innovation $(r_k^N)_{k \geq 1}$ after t_0 is unknown, it is necessary to detect both negative and positive deviations of the mean innovation $\mathbb{E}_{t_0}^1(r_k^N)$ from the target value 0. The stopping time t_a^{TS} of the two-sided *CUSUM* test is given by [15] :

$$\begin{aligned} t_a^{TS} &= \inf \{k \geq 1 : (g_k^{upp} \geq h) \vee (g_k^{low} \geq h)\} \\ g_k^{upp} &= \left(g_{k-1}^{upp} + \frac{\nu}{\sigma^2} \left(u_k - \frac{\nu}{2} \right) \right)^+ \\ g_k^{low} &= \left(g_{k-1}^{low} + \frac{\nu}{\sigma^2} \left(-u_k - \frac{\nu}{2} \right) \right)^+ \end{aligned} \quad (18)$$

where $g_0^{upp} = g_0^{low} = 0$.

²Let us assume that $y, \hat{y}, x$ are the random values. We say that the $y = \text{esssup } x$ if : 1) $\mathbf{P}(x \leq y) = 1$; 2) if $\mathbf{P}(x \leq \hat{y}) = 1$ then $\mathbf{P}(y \leq \hat{y}) = 1$, where $\mathbf{P}(A)$ is the probability of the event A .

B. Automatic tuning of the CUSUM parameters

Understanding that the tuning parameters h and ν are crucially important to define the mean time before a false alarm γ and the worst case mean detection delay $\bar{\mathbb{E}}^*(t_a)$, it is necessary to find a tuning methodology that insures the suitable performances of *CUSUM* test.

First of all, let us define how to compute γ and $\bar{\mathbb{E}}^*(t_a)$ as functions of tuning parameters h and ν . We briefly describe now an exact computation of these criteria based upon the solution of some Fredholm integral equations (see details in [15]). To define the performance of a statistical test, the power function is usually used. In the sequential change detection problem, the analog of this function is the average run length (ARL) function :

$$\text{ARL}(m) = \mathbb{E}_m(t_a). \quad (19)$$

Hence, it is assumed that all observations $(u_k)_{k \geq 1}$ follow the same Gaussian distribution with the mean $m = \mathbb{E}(u_k)$. It can be shown that $\bar{\mathbb{E}}^*(t_a) = \text{ARL}(\nu)$ and $\gamma = \text{ARL}(0)$. It can be also shown that (see details in [15])

$$\text{ARL}(\theta) = \frac{N_\theta(0)}{1 - \mathbf{P}_\theta(0)}, \quad (20)$$

where θ is the parameter defining the distribution of the *CUSUM* log likelihood ratio (LLR) for one observation

$$Z_k = \frac{\nu}{\sigma^2} \left(u_k - \frac{\nu}{2} \right)$$

and the functions $z \mapsto N_\theta(z)$ and $z \mapsto \mathbf{P}_\theta(z)$ are defined by the following Fredholm integral equations of the second kind :

$$\mathbf{P}_\theta(z) = \int_{-\infty}^{-z} f_\theta(x) dx + \int_0^h \mathbf{P}_\theta(x) f_\theta(x - z) dx \quad (21)$$

$$N_\theta(z) = 1 + \int_0^h N_\theta(x) f_\theta(x - z) dx \quad (22)$$

for $0 \leq z \leq h$, where f_θ is called the kernel of the integral equation. In the framework of sequential change detection problem, the kernel $f_\theta(x)$ represents the probability density of the LLR. In our case this density corresponds (at least approximately) to a Gaussian law with the mean $\mathbb{E}(Z_k)$ and the variance $\text{Var}(Z_k)$. Hence, the parameter vector is defined as $\theta = (\mathbb{E}(Z_k), \text{Var}(Z_k))^T$. Finally, the elements of θ are expressed as functions of two first moments of the random sequence $(u_k)_{k \geq 1}$:

$$\mathbb{E}(Z_k) = \frac{\nu}{\sigma^2} (m - \frac{\nu}{2}) \quad (23)$$

$$\text{Var}(Z_k) = \frac{\nu^2}{4\sigma^2}. \quad (24)$$

For solving numerically equations (21) - (22), two approaches have been proposed : i) to solve the integral equation by using an iterative method; ii) to replace the integral equation by a system of linear algebraic equations and to solve this system with respect to the unknown variables. Nowadays, the second method is mainly used.

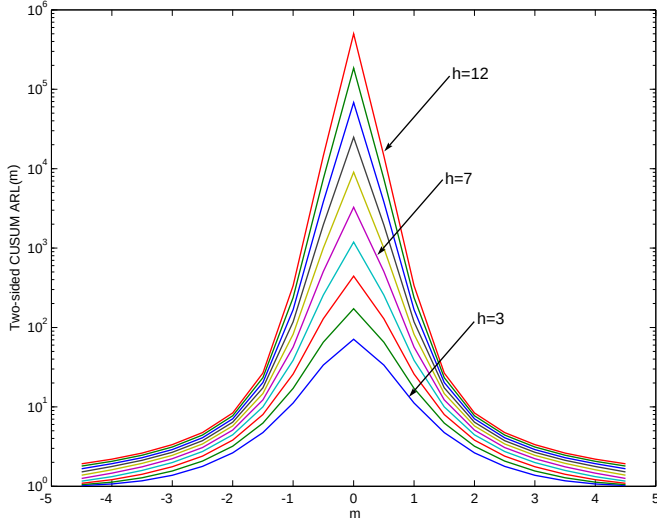


Fig. 1. The ARL functions $m \mapsto \text{ARL}(m)$ for different thresholds $h = \{3, 4, \dots, 12\}$.

The ARL function of symmetric two-sided *CUSUM* test can be expressed in term of ARL functions defined for one-sided *CUSUM* (see details in [15]) :

$$\frac{1}{\text{ARL}_{TS}(\theta)} = \frac{1}{\text{ARL}_l(\theta)} + \frac{1}{\text{ARL}_u(\theta)}, \quad (25)$$

where $\text{ARL}_u(\theta)$ (resp. $\text{ARL}_l(\theta)$) is the ARL function of one-sided *CUSUM* with the decision function g_k^{upp} (resp. g_k^{low}) defined in equation (18).

To illustrate the above mentioned equations, let us assume that the following parameters of two-sided *CUSUM* test (18) are chosen : $\sigma^2 = 1$, $\nu = 3$. Let us assume that the true expectation of u_k is defined by $m = \mathbb{E}(u_k)$. First of all, let us compute the ARL functions $m \mapsto \text{ARL}(m; h)$ for different values of threshold $h = \{3, 4, \dots, 12\}$. The results of computation are presented in Figure 1. It is easy to see that the mean time before a false alarm γ is heavily based on the value of h . The function $h \mapsto \text{ARL}(0; h)$ is rapidly increasing. It is obvious that the mean detection delay $h \mapsto \text{ARL}(m; h)$ for $|m| \geq \nu$ is also increasing but as an almost linear function.

Second, let us check now the sensitivity of the two-sided *CUSUM* to the true variance of the random process $(u_k)_{k \geq 1}$. Let us define the ratio $R = \sigma_t^2 / \sigma^2$, where σ_t^2 is the true variance of the random process $(u_k)_{k \geq 1}$ and the σ^2 is its assumed value. The results of computation are presented in Figure 2. Here, the ARL functions $m \mapsto \text{ARL}(m; R)$ are computed for different values of ratio $R = \{0.5, 0.6, \dots, 1.5\}$. It is easy to see that the mean time before a false alarm is very sensitive with respect to the ratio R . The function $R \mapsto \text{ARL}(0; R)$ is rapidly decreasing. On the contrary, the mean detection delay is not very sensitive with respect to R , i.e. the function $R \mapsto \text{ARL}(m; R)$ for $|m| \geq \nu$ is almost constant.

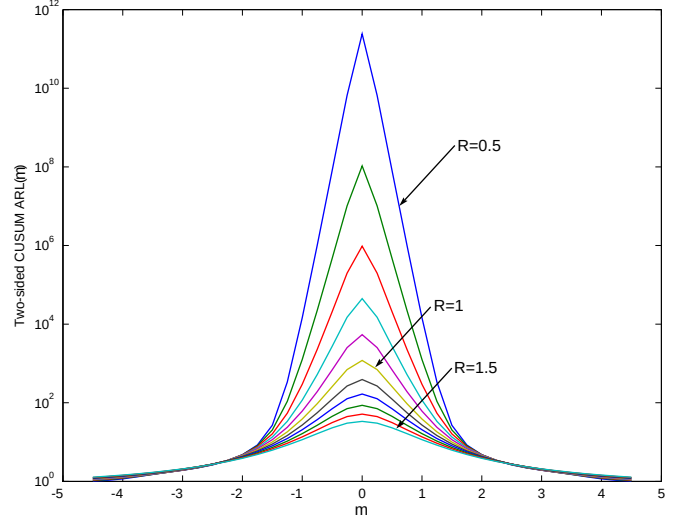


Fig. 2. The ARL functions $m \mapsto \text{ARL}(m)$ for different ratios $R = \{0.5, 0.6, \dots, 1.5\}$.

IV. PERFORMANCE EVALUATION

In this section, we briefly recall the basics of inertial navigation update before assessing the performance of the two-sided *CUSUM* algorithm applied to divergence detection.

An aircraft equipped with an *IMU*³ is able to estimate its geographical position as well as its speed by integrating the gyroscope and the accelerometer measurements. However, sensor errors are responsible for a drift in these estimates. By using radar-altimeter measurements, it is possible to update the position and velocity.

Let $X^{\text{geo}} = (L \ l \ z \ v^n \ v^e \ v^d)^T$ be the vector containing the unknown true position and velocity of the aircraft. L is the latitude, l the longitude and z the altitude in the navigation frame. The components v^n , v^e and v^d denote the velocity in the north, east and down directions. Given that $\hat{X}^{\text{geo}}$ is the position given by the *IMU*, we can define the error vector in the navigation frame $x = \delta X$ between X^{geo} and $\hat{X}^{\text{geo}}$ as $\delta X = (\delta X^n \ \delta X^e \ \delta X^d \ \delta v^n \ \delta v^e \ \delta v^d)^T$. The navigation filter tries to estimate $x_k = \delta X_k$ via the following dynamical and measurement equations:

$$\begin{cases} x_{k+1} = F x_k + w_k \\ y_k = h(x_k) + v_k \end{cases} \quad (26)$$

where $F = \begin{pmatrix} I_3 & \Delta I_3 \\ 0_3 & I_3 \end{pmatrix}$, $w_k \sim \mathcal{N}(0, Q)$ and $v_k \sim \mathcal{N}(0, \sigma_v^2)$. The measurement function h has the following expression:

$$h(x_k) = \tilde{z}_k + x_k^3 - \text{DTED}(X_k^L, X_k^l)$$

where $X_k^L = \tilde{L} + \frac{x_k^1}{R^n + \tilde{z}_k}$ and $X_k^l = \tilde{l} + \frac{x_k^2}{(R^e + \tilde{z}_k) \cos(\tilde{L})}$.

³Inertial Measurement Unit

R^n and R^e represent respectively the curvature radii in the north direction and east direction.

The *DTED* (*Digital Terrain Elevation Data*) maps a longitude and a latitude component to the corresponding terrain height.

A. Performance indicators for the CUSUM test

In the sequel, the *CUSUM* test (section III) is used as follows. We focus on the mean detection delay instead of the worst detection delay (15). Moreover, instead of using the mean time between false alarm, the false alarm rate is a more suitable divergence indicator for our application.

To assess the accuracy of the two-sided *CUSUM* test, we look at the following indicators:

- the *false alarm rate* r_{FA} : we define the false alarm rate as the rate at which the two-sided *CUSUM* test declares a divergence a time $t_a^{TS} < t_0$, t_0 being the true time of divergence (with $t_0 = \infty$ when the filter has not diverged during the whole path).
- the *non detection rate* r_{ND} : it is the rate at which the detection algorithm fails to raise an alarm when the filter has actually diverged
- the *mean detection delay* $\tau = \mathbb{E}(t_a^{TS} - t_0 + 1 | t_a^{TS} \geq t_0)$

To compute the three indicators, we consider a baseline flight path for the aircraft corresponding to the true position of the aircraft which will remain fixed. A number $N_{MC} = 250$ of inertial positions are then simulated according to the model in [16]. For each of these positions, a particle filter provides an estimate $\hat{X}_k^{geo}, k = 1, \dots, n$ of the position and velocity vector, n being the total number of measurements.

Let N_{FA} be the number of false alarms, N_{ND} the number of non-detected divergent tracks and N_{TD} the number of true divergences. Then r_{FA} and r_{ND} are computed as follows:

$$r_{FA} = \frac{N_{FA}}{N_{MC}}$$

$$r_{ND} = \frac{N_{ND}}{N_{TD}}$$

The true time of divergence t_0 must be evaluated automatically for each filter run in order to derive r_{FA} , r_{ND} and τ . Assuming that the particle cloud is unimodal, this can be done according to the following criterion: the filter has diverged if the true position is outside the $100(1 - \beta)\%$ confidence ellipsoid given by the empirical covariance $\hat{\Sigma}_k$ for 5 consecutive time steps. β should be between zero and one, in our case we used an empirically chosen value $\beta = 0.05$. However, this criterion is inefficient for multimodal distributions since confidence ellipsoids only make sense for unimodal distributions. One solution is to use a mixture particle filter [3], [4] which decomposes the particle filter $\hat{p}_k^N$ into several unimodal components $\hat{p}_j(x_k|y_{0:k}), j = 1 \dots M$ such that:

$$\hat{p}_k^N = \sum_{j=1}^M \alpha_j \hat{p}_j(x_k|y_{0:k})$$

where $\sum_{j=1}^M \alpha_j = 1$ and $\alpha_j \geq 0, \forall j$.

Each component is represented by a cluster of particles. To check for filter divergence, we compute the empirical covariance matrix $\hat{\Sigma}_k^j$ of each cluster and determine if the true state lies in $\hat{\Sigma}_k^j$'s confidence ellipsoid for a given level. Figure 3 provides an illustration of this method.

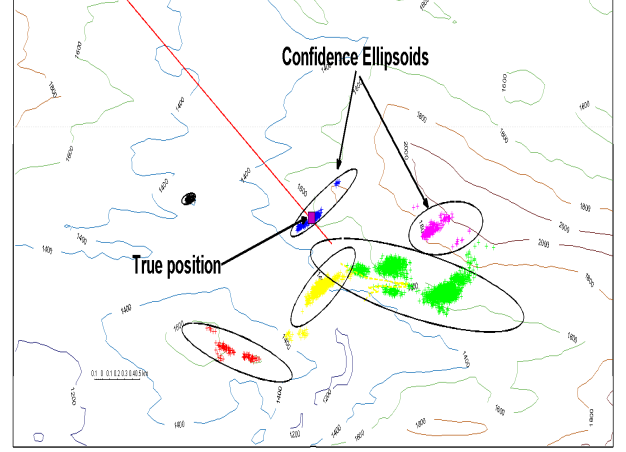


Fig. 3. Particle clustering and confidence ellipsoids. Each cluster is represented by a different colour

B. Discussion of numerical results

To fully evaluate the abilities of the divergence detector, we consider three kinds of terrains which we will refer to as terrain A, B and C.

- terrain A has low to medium ambiguity
- terrain B has medium ambiguity
- terrain C has high ambiguity

The ambiguity is characterized by the similarities of some regions of the terrain. In the early stages of a filter implementation, this is evidenced by the presence of several clusters of particles: to one radar-altimeter measurement y_k corresponds several separate positions in the horizontal plane. A typical example is given in figure 3 (Terrain B).

This has an immediate impact on the efficiency of the divergence detector based on the normalized innovations $\{r_k^N\}$. In the case of a highly ambiguous terrain, the predicted measurement $\hat{y}_{k|k-1}$ may actually match the ground clearance y_k even though the particle cloud has diverged. In this case, the divergence does not correspond to a change in the mean of r_k^N . This implies that the two-sided *CUSUM* test will only detect a problem much later, when there is an actual mismatch between the received measurement and the 'particle measurements' (see figure 5). Therefore the actual detection delay is the sum of the intrinsic delay of the two-sided *CUSUM* algorithm and the delay caused by the terrain ambiguity. An ideal case is illustrated by figure 4 where the ambiguity is low and the detection is quick.

In the experimental setup we assume that the initial uncertainty on the position and velocity error components follows a

Gaussian distribution $x_0 \sim \mathcal{N}(0, \Sigma_0)$. The following values have been used in our simulations:

$$\begin{cases} Q = \text{diag}(0\text{m}, 0\text{m}, 0\text{m}, 0.1\text{m/s}, 0.1\text{m/s}, 0.01\text{m/s})^2 \\ \sigma_v = 10\text{m} \\ \Sigma_0 = \text{diag}(1\text{km}, 1\text{km}, 100\text{m}, 3\text{m/s}, 3\text{m/s}, 2\text{m/s})^2 \\ \Delta = 0.1\text{s} \\ n = 1000 \text{ filter iterations} \end{cases}$$

The value of the measurement noise standard deviation σ_v is quite small with respect to the terrain variation so that the likelihood and the prior will often have poor overlap. This implies high Monte Carlo error which leads to a high divergence rate [12]. The simulations were carried out using a mixture regularized particle filter as in [3]. Moreover, the filter has not been carefully initialized so as to increase chances of observing divergent tracks.

TABLE I
FALSE ALARM RATE $r_{FA}(\%)$

	Ter. A	Ter. B	Ter. C
$h = 8$	6	5.2	6.8
$h = 9$	3.6	2.4	1.6
$h = 10$	2	1.2	0.4
$h = 11$	1.2	0.8	0.4
$h = 12$	0.8	0.4	0
$h = 13$	0.4	0.4	0

TABLE II
MEAN DETECTION DELAY τ

	Ter. A	Ter. B	Ter. C
$h = 8$	19	41	140
$h = 9$	19	43	166
$h = 10$	20	43	189
$h = 11$	21	43	193
$h = 12$	22	43	205
$h = 13$	22	44	223

TABLE III
NON DETECTION RATE $r_{ND}(\%)$

	Ter. A	Ter. B	Ter. C
$h = 8$	2.4	3.2	3.2
$h = 9$	2.8	3.6	4.4
$h = 10$	3.6	5.2	4.4
$h = 11$	4	5.6	5.2
$h = 12$	4.4	6.8	5.6
$h = 13$	5.2	6.8	6.8

The three indicators r_{FA} , r_{ND} and τ have been computed for several values of the two-sided *CUSUM* threshold h considering a fixed mean shift $\nu = 3$. They are displayed in tables I, II and III. As stated in section III-B (fig. 1), the

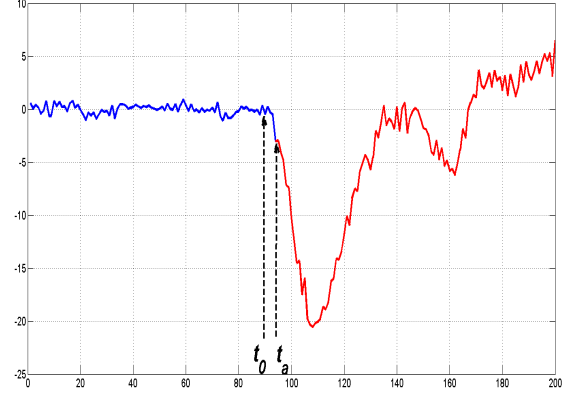


Fig. 4. Normalized innovation in a case of divergence (low terrain ambiguity, first 200 iterations) Red: after divergence detection

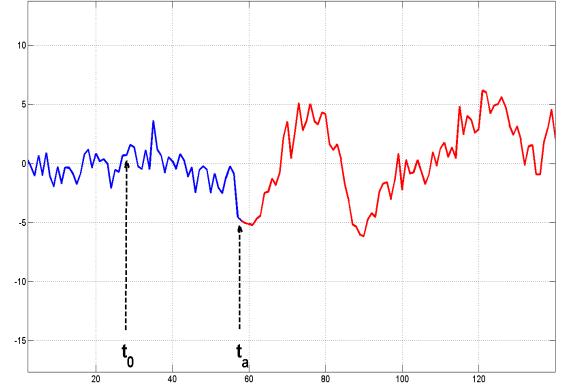


Fig. 5. Normalized innovation in a case of divergence (high terrain ambiguity, first 140 iterations) Red: after divergence detection

false alarm rate r_{FA} is much more sensitive to h than the mean delay of detection r_{ND} . Because we favor a low false alarm rather than a low detection delay, high values of h are advisable (r_{FA} is less than 1%). Indeed, it would be costly to restart a filter that has not actually diverged whereas we can tolerate a delayed diagnosis of divergence.

As expected, the nature of the terrain has a big impact on the detection delay. When the terrain has low to medium ambiguity, the mean detection delay is around 19 to 44 time steps (for a total number of $n = 1000$ filter iterations), which is acceptable if the goal is to restart the filter once divergence is detected. However, as can be seen in table II, the mean detection delay increases noticeably to over 100 iterations for the most ambiguous terrain.

Finally, the non detection rate increases much slower with h than the false alarm rate decreases. For instance, for terrain A, r_{FA} is divided by 15 with a threshold of $h = 13$ instead of $h = 8$, whereas r_{ND} is only doubled with this increase of h . Once divergence has been detected, a possibility can be to reset the filter by considering a vague prior represented for

instance by a large covariance matrix.

V. CONCLUSION

We have introduced a framework for monitoring particle filter divergence in a sequential manner. The methodology was developed assuming scalar measurements as is the case in navigation using a radar-altimeter, but it can be extended to the multidimensional case. The divergence detection is based on the two-sided *CUSUM* algorithm and uses the normalized innovation as a statistics. Its efficiency is demonstrated in the case where the terrain is not too ambiguous. More precisely, on two test cases, it is possible to achieve low false alarm rates with reasonable mean detection delay. In the case of high ambiguity, the detection delay increases considerably since a change in the mean of the innovation is not sufficient to characterize filter divergence.

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